CorpusGuard

AI Security Assessment Report

Target System	RBC AML Compliance Agent (Simulation)
Assessment Date	July 23, 2026 — 23:54 UTC
Framework Version	CorpusGuard 0.1.0
Research Reference	SSRN 6734225 — Ondieki Ombachi, 2026
Assessment Type	RAG Corpus Poisoning Security Assessment

Executive Summary

Risk Score	**85/100 — CRITICAL**
Vulnerabilities Found	2
Attack Vectors Tested	3 (QTPI, Document Poisoning, Rank Manipulation)
MHL Detection Rate	100% (0% false quarantine) — published, SSRN 6734225
Regulatory Frameworks Affected	NIST AI RMF, OWASP LLM Top 10, FATF 2025

This assessment evaluated the target system's resilience against adversarial corpus poisoning attacks — a class of attack first formally documented in SSRN 6734225 (Ondieki Ombachi, 2026). These attacks exploit the retrieval corpus layer of RAG-based AI systems, injecting adversarial documents through legitimate ingestion pathways to collapse system accuracy while generating zero anomalous log signatures.

Attack Results

Attack Vector	β (docs)	Baseline F1	Attacked F1	Degradation	Log Anomalies
QTPI (Prompt Injection)	50	0.919	0.014	**98.4%**	0
Document Poisoning †	280	0.919	0.924	+0.5%	0
Rank Manipulation †	280	0.919	0.943	+2.6%	0

QTPI achieves 98.4% F1 degradation at the tested injection budget. The agent becomes operationally equivalent to random guessing while producing no anomalous log signatures. † Document Poisoning and Rank Manipulation rows are published reference values from SSRN 6734225 (this assessment ran the QTPI vector); the paper reports QTPI collapsing F1 from 0.919 to 0.014 (98.4% degradation) at β=50.

Defense Recommendations

Deploy the Memory Hygiene Layer (MHL) as a pre-ingestion gate in your RAG pipeline. No model retraining required. No infrastructure changes required.

Priority	Action	Effort	Impact
1 — Critical	Deploy CPT (Cryptographic Provenance Tracking)	2 hours	100% QTPI detection
2 — High	Configure FCS threshold at 0.65	1 hour	Reduces FP rate to 0%
3 — High	Enable SRAD sliding window monitoring	1 hour	Corpus-level anomaly detection
4 — Medium	Integrate MHL into document ingestion pipeline	1 day	Full protection
5 — Medium	Set up Prometheus metrics + alerting	2 hours	Operational visibility

Regulatory Mapping

Framework	Gap Identified	MHL Mitigation
NIST AI RMF	No profile for retrieval corpus integrity	First reference implementation
OWASP LLM Top 10	Corpus-level prompt injection not covered	QTPI formalises new LLM01 sub-category
MITRE ATLAS	No technique for corpus manipulation	New attack pattern documented
FINRA Rule 3110	QTPI leaves no audit signature	CPT provides cryptographic audit trail
FinCEN 2024	Agent at F1=0.014 not reasonably designed	CPT + SRAD provide auditable controls
FATF 2025	LLM-generated docs bypass human review	CPT detects source-level anomalies

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